Dental Practice Transition Brokerage Platform (India-first): Research & PRD

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1. Executive Summary

Build a broker-led transaction system for Indian dental practice sales, with a controlled-access listing front end, and launch it as an 18-week MVP that your own brokerage team uses on real deals. This PRD is built on the three answers you gave: India first, your brokerage team runs the deals, dental only. Research date: 7 November 2026.

What the research found

- Of ten competitors, only Dental Acquisitions shows masked listings, an NDA gate and a deal room on its public pages, and it operates only in the US. The only India-focused dental player I found, SureMedico, runs equipment and clinic classifieds with no financial disclosure or deal tooling (section 4).
- India's dental market is large and fragmented, but I found no verifiable transaction volumes or valuation multiples, so the market opportunity is a hypothesis to test with interviews (section 3).
- One competitor, Practice Buy Sell, publishes placeholder testimonials and a placeholder phone number; its claims are excluded from benchmarks.

What is recommended

Decision	Recommendation	Section
Positioning	Broker-led hybrid: anonymised listings feed an internal CRM and deal workspace; not an open marketplace	6
MVP	35 features, 31 of them Must-haves; confidentiality enforced on the server in four disclosure tiers	13, 21
Stack	Next.js, Express, MongoDB as mandated, plus a shared Zod package, private object storage and a job queue	18
AI	None in the MVP; deterministic matching and valuation, with AI limited to drafting and document reading later	20

Decision	Recommendation	Section
Valuation	Deferred to Phase 2 until multiples are validated with real Indian deals	10E
Timeline	About 18 weeks with a team of six (planning estimate)	25

What must happen before build starts

- Eight founder decisions, including launch cities, fee model and whether your team represents both sides (section 27).
- Legal review of dental practice ownership, licence transfer, advertising rules, the NDA text and DPDP compliance (section 19). The DPDP Rules are scheduled to apply fully from 13 May 2027, and I could not confirm whether that date has moved.

Limits of this document. Competitor evidence comes from public homepages; nothing behind a login was inspected. Targets, fee-related figures and retention periods are labelled proposals. Nothing here is legal, tax or financial advice.

2. Problem Statement

Indian dental owners who want to retire, exit or bring in a partner have no confidential, structured way to find qualified buyers, prove their numbers and run the deal; the brokerage team that serves them has no system to run that work. The pain points below come from your brief and from how US competitors position themselves; they are **hypotheses until the interviews recommended in section 3 confirm them for India**.

Participant	Pain point	How the platform responds
Seller (owner, retiring or exiting)	Cannot find trustworthy buyers without telling staff and patients	Anonymised listings; named identity only after NDA and broker approval
Seller	Does not know what the practice is worth	Transparent preliminary estimate, clearly not a certified valuation
Seller	Financial records are scattered or informal	Guided financial intake and a categorised document checklist
Seller	Ownership transfer, lease and staff handover are unclear	Transaction stages with an outstanding-requirements list through handover

Participant	Pain point	How the platform responds
Buyer (associate, neighbouring dentist, group)	Cannot discover suitable practices	Search, filters, saved searches and broker-curated matches
Buyer	Cannot verify revenue, assets or risk before committing time	Staged disclosure and a verified-document data room
Buyer	Needs financing and does not know what to prepare	Buyer qualification step and a financing-readiness checklist (lender integration deferred)
Broker	Leads, documents and conversations live in phones and spreadsheets	Broker CRM: leads, tasks, notes, communication history
Broker	Cannot see where each deal is stuck	Kanban pipeline with configurable stages and aging alerts
Broker	Coordinating NDAs, documents and offers by email	One workspace per deal with access logs and offer history

What this platform is not. It is not a classified-ads site and does not store patient treatment records. Patient-identifiable data stays out of the MVP, which keeps it a business-transfer tool rather than a clinical system.

3. Market Research

India's dental practice market is large and fragmented, but I found no verifiable transaction data for it, so the opportunity below is a hypothesis to validate, not a sized market. Research date: 7 November 2026.

What the sources support

- **Fragmentation.** Older press on Investcorp's investment in Global Dental Services said hospitals and chains were under 10% of practices and valued Indian dental services at about $3 billion; both figures are several years old.
- **Dentist counts conflict.** One industry interview cites 180,000 dentists and 125,000 clinics; a clinic-chain blog says about 300,000 dentists. Neither is an official register count.

- **Chains own, not acquire.** The same interview says Clove Dental, Sabka Dentist and Apollo fully own their centres, and a chain profile describes 112 company-owned, company-operated clinics for Sabka. A trade report said Clove was considering a controlling stake in Sabka. Institutional demand for independent practices is therefore unproven in India.
- **Visible supply is informal.** SureMedico's marketplace lists mostly equipment, chair-sharing and clinic setups, with clinic prices from roughly ₹1.5 lakh to ₹17 lakh marked "neg.".

India versus the US reference market

Dimension	India (what I could verify)	US (competitor evidence)
Typical listing	Clinic setups and equipment on rent or sale; few financial fields	Practices listed with price, operatories and gross collections, e.g. National Practice Sales
Specialist intermediaries	None found beyond SureMedico's classifieds and SMERGERS' generalist network	Dozens of brokerages; 42+ indexed by The Practice Market
Buyer pool	Associates and neighbouring dentists likely; chain demand unproven	Associates, group practices and corporate buyers per Henry Schein DPT
Privacy law	DPDP Act and Rules (see section 19)	HIPAA only where patient data is involved; state privacy laws
Advertising rules	Dentists' code of ethics restricts advertising (DCI Code of Ethics); compliance route for a listing platform needs counsel	Varies by state

Where the opportunity most likely sits (hypothesis)

The best-supported gap is a confidential, broker-managed transition service for solo and small-group dental practices in metros: retiring owners, partner buy-outs and associate-to-owner sales. Evidence for it: nobody in India appears to offer structured financial disclosure, NDAs and a deal room for dental practices, while the US shows these are what serious brokers and marketplaces compete on.

Cannot be verified from public sources (validate before launch)

- Number of dental practice transactions per year in India, and typical deal sizes.

- Valuation methods and multiples in use; I give no multiple here because I found no reliable Indian source.
- Quality of sellers' financial records and the share of income that is documented.
- Appetite of chains and investors to buy independent practices.

Recommended validation (my proposal): 15–20 interviews with owners aged 55+, associates, and two chains' corporate development teams, plus collection of 10 real recent transactions, before finalising pricing and the valuation module.

4. Competitor Analysis

No competitor combines India-specific dental practice data, confidential listings, a data room and a broker CRM; the closest India-based player, SureMedico, is an equipment-and-clinic classifieds site. Research date: 7 November 2026. I opened each competitor's homepage (plus SureMedico's marketplace page) with a live fetch; nothing behind a login was inspected.

Evidence labels. V = stated or visible on the public page I opened. **A** = a marketing claim I could not verify (counts, revenue, outcomes). **U** = not visible without login or not rendered in the fetch, so unknown. Anything in section 6 onward labelled "Recommendation" is my own proposal, not a competitor feature.

Platform	Model and target	Verified on public pages (V)	Advertised, unverified (A)	Gaps and notes
VetValue	Founder-led brokerage and business platform for independent US veterinary owners	Three tracks (Grow, Connect, Share); a free-valuation page; gated member app login; private forum under real names; a Pricing page for membership	$1.3B+ transactions; sale fees 45–70% below traditional brokers	Veterinary only; no public listings seen; the sale workflow is described, not shown

Platform	Model and target	Verified on public pages (V)	Advertised, unverified (A)	Gaps and notes
Henry Schein Dental Practice Transitions	Corporate dental brokerage run by territory-based consultants	Coded blind listings (state, type, gross collections, listing code); a Preferred Purchaser form to see full listings; an NDA form; valuation and EBITDA enquiry forms; loan calculator; consultant directory; state-level valuation and sales table; sign-in	36,000 purchasers; 2,000 valuations; $500M+ in sales; never represents both sides	Broker-led, so no visible self-serve deal room; US only (brokering excludes Rhode Island, per its FAQ)
The Practice Market	Free aggregator indexing broker-published listings across 19 professions	No-login browsing; every listing links to the representing broker; seller-to-broker matching, no for-sale-by-owner; claims neutral, non-paid ranking	5,400 listings, 3,844 of them dental	Pure discovery layer: no NDA, documents or deal tracking; US data sourced from public broker sites
Clinic Exchange	Peer-to-peer US dental marketplace	Only the tagline and metadata rendered: "confidential listings" and "structured data rooms"	None checkable	**U**: workflows, pricing and verification are unknown from the public fetch

Platform	Model and target	Verified on public pages (V)	Advertised, unverified (A)	Gaps and notes
Practice Transitions Group	Sell-side M&A advisory for dental, medical, med spa, dermatology, behavioural health and veterinary practices ($2–30M revenue)	Per-specialty valuation calculators; public listings; buyer-list signup; case studies; claims buyers see details only after vetting	140+ owners served; 23% average price increase; 3–5 qualified offers per sale	Advisory-led, US; no platform workflow shown
SMERGERS	Horizontal SME deal marketplace, run by an Indian company (SMERGERS Online Services Private Limited); sells to buyers, investors, advisors and lenders globally	Profiles for businesses, investors, advisors and franchises; a valuation calculator and a CIM (information memorandum) builder page; a "Clinics for sale" category; card, net-banking and Stripe payments; refund policy; stated selective identity disclosure	74,000+ pre-screened profiles (its own footer says 20K businesses and 80K investors, so the figures conflict)	Not dental-specific: no operatories, production or payer-mix fields; no visible deal room

Platform	Model and target	Verified on public pages (V)	Advertised, unverified (A)	Gaps and notes
SureMedico	Free Indian dental professional network with a buy/sell marketplace	Four listing kinds (item, material lot, whole-clinic setup, practice for sale); admin-approved listings; verified doctors only; seller name and phone hidden until 1:1 connect; INR prices; many clinic-for-rent and chair-sharing listings	None material	Closest India comparable, but it is classifieds: no financial disclosure, NDA, valuation or deal pipeline visible; one "practice" is listed at ₹1.5 lakh ("neg.")
National Practice Sales	California brokerage for dental, medical, veterinary and optometry practices	Listings with price, operatories, area and 2025 gross income; full financials released after NDA; a five-stage seller process; written opinion of value in five business days; financing page; a "Recently sold" page	100+ transitions closed; 90–180 day typical timeline	One state; broker-led; no platform workflow shown

Platform	Model and target	Verified on public pages (V)	Advertised, unverified (A)	Gaps and notes
Dental Acquisitions	US dental marketplace that says it is not a broker; adds a Deal Room for active transactions	Find, Qualify, Close flow; listings masked to an area, not an address, until a buyer qualifies and signs an NDA; sellers can list directly or co-manage with a broker; brokers list free and pay a fee at closing; attorneys, CPAs and lenders can be invited; valuation calculator; 3,141 listings	No pay-for-visibility claim	US only (Canada in beta); the page labels its workflow visuals "illustrative product placeholders"
Practice Buy Sell	Confidential brokerage for medical, dental and veterinary practices	A lead-capture page with a valuation request form	$500M+ transactions; 600+ sold; 95%+ close rate	**Do not benchmark.** Its testimonials are literally labelled "Placeholder — replace with a real review" and its phone number is a 555 placeholder; treat all its claims as unverified

What this means for the product

- The two platforms with the strongest transaction design, Dental Acquisitions (masked listings, Deal Room, invited professionals) and Clinic Exchange (data rooms), are both US-only, so India has no equivalent.

- India's visible dental activity is equipment and chair-sharing classifieds (SureMedico) and horizontal SME listings (SMERGERS). Neither captures dental metrics such as chairs, patient volumes or revenue mix.
- Every broker-led site gates full financials behind an NDA and a vetted-buyer step; the platform should copy that pattern, not invent a looser one.
- None of the pages I opened shows a public, off-the-shelf broker CRM, so the broker-facing tooling is a gap I could not confirm any competitor fills. Verify this with a demo request before treating it as a certain differentiator.

5. Competitive Feature Matrix

Only Dental Acquisitions shows a masked-listing, NDA and deal-room flow on its public pages, and no competitor shows a broker CRM; the planned platform's row shows where it would match or lead. Research date: 7 November 2026; homepages only, so "–" means *not seen*, not *absent*.

Legend: V = stated or visible on a public page · **A** = marketing claim only · **?** = page did not render the detail · **–** = not seen. Plan column uses MVP / P2 / P3 for release phase (sections 21–22).

Platform	Public listings	Masked or blind listings	NDA before financials	Valuation tool	Data room / diligence	Deal workspace	Broker CRM	India dental data
VetValue	–	A	–	V (free valuation page)	?	A ("sale management platform")	–	–
Henry Schein DPT	V	V (listing codes, city only)	V (NDA and purchaser forms)	V (valuation and EBITDA enquiry forms)	?	?	?	–
The Practice Market	V (aggregated)	–	–	–	–	–	–	–
Clinic Exchange	?	A	?	?	A ("structured data rooms")	?	?	–
Practice Transitions Group	V	A	A	V (calculators)	?	A	?	–
SMERGERS	V	A (selective disclosure)	–	V (calculator)	–	–	–	Partial (Indian company; generalist)
SureMedico	V	V (seller identity hidden)	–	–	–	–	–	V

Platform	Public listings	Masked or blind listings	NDA before financials	Valuation tool	Data room / diligence	Deal workspace	Broker CRM	India dental data
National Practice Sales	V	V (stated)	V (stated)	V (opinion in 5 days)	?	?	?	–
Dental Acquisitions	V	V (area-level map)	V (stated)	V (calculator)	V (Deal Room, invited professionals)	V (Deal Room)	–	–
Practice Buy Sell	–	A	A	V (request form only)	A	A	–	–
Planned platform	MVP (authorised buyers only)	MVP	MVP (clickwrap; e-sign in P3)	P2 (preliminary estimate)	MVP (uploads) / P2 (checklists)	MVP	MVP	MVP (dental, India)

Reading the matrix

- NDA-gated financials is table stakes: every broker site that mentions disclosure uses it, so the MVP must have it.
- A deal room is the differentiator among US marketplaces; building it for India dental is the strongest defensible position.
- The broker CRM is the gap in competitor evidence, but internal tooling is invisible on public pages, so treat it as an assumption until a demo or broker interview confirms it.

6. Product Vision

Build a broker-led M&A transaction system for Indian dental practices, with a controlled-access listing front end, rather than an open marketplace. This follows your answer that your own brokerage team runs the deals, and it matches the competitor evidence: the strongest products pair confidential discovery with a managed deal workspace.

Element	Statement
Mission	Help Indian dental owners exit, retire or bring in partners confidentially and at a fair price, and help qualified dentists buy with verified information.
Vision	The default system of record for dental practice transitions in India, where every deal from first enquiry to post-sale handover is tracked in one place.

Element	Statement
Value proposition	Sellers keep control of who learns what and when; buyers see verified information in stages; your brokers run every deal from one CRM and workspace.
Target audience (MVP)	Solo and small-group dental practice owners and buyers in your launch cities (**city list to be confirmed**), plus your internal brokerage team.
Core problems solved	Confidentiality, buyer discovery, financial verification, deal tracking, handover (section 2).
Differentiation	India-dental specificity; staged disclosure with access logs; internal CRM tied to the deal workspace; transparent preliminary valuation.

Why a broker-led hybrid, not the alternatives

Option	Verdict	Reason
Open marketplace	Not for MVP	Needs two-sided liquidity you do not have; classifieds sites like SureMedico already own the casual end.
Pure broker CRM	Too narrow	Gives brokers tools but does not generate buyer or seller demand.
Transaction management system only	Partial	Strong for diligence and closing, weak at discovery.
Broker-led hybrid (recommended)	**Adopt**	Listings generate demand, the CRM and deal workspace deliver the service, and every listing is approved by your team.

External brokers are out of scope for the MVP and are a Phase 3 option. **Founder decision needed:** confirm that your brokerage team is the only party that can create and approve listings at launch.

Primary success metrics (targets to be set by you): approved listings; buyers qualified per listing; days from NDA to offer; days from first enquiry to close; share of deals tracked end-to-end in the system.

7. Business Model

The recommended default is a success fee paid at closing by the seller, with optional paid services layered on later; no fee levels are set here because you have not confirmed any. Competitor practice, where visible, points the same way.

Revenue option	Payer	Phase	Competitor evidence	Status
Success fee at closing	Seller (default)	MVP	Dental Acquisitions charges brokers a fee at close; Practice Buy Sell claims success-based fees (unreliable source)	Rate **to be set by founder**
Listing or retainer fee	Seller	Optional	Not seen publicly	Founder decision
Paid valuation report	Seller	P2	Free valuation tools are common (Henry Schein, PTG, Dental Acquisitions), so charging may hurt conversion	Needs testing
Buyer premium access	Buyer	P2	Henry Schein and The Practice Market keep buyer access free	Likely keep free
Lender, legal or accounting referral fees	Third party	P3	Not seen	Needs legal review for conflicts

Policy decisions only you can make

- **Representation.** Henry Schein says it never represents both sides, and PTG says it never takes buyer fees (both marketing claims). Decide whether your team represents sellers only or both parties; the answer changes permissions, fee rules and conflict disclosures.
- **Fee basis.** Percentage of deal value, tiered percentage, or fixed fee.
- **Tax and invoicing.** Confirm GST treatment of brokerage fees and invoicing rules with an accountant.

Business success metrics to instrument from day one: listings approved, buyer-to-NDA conversion, NDA-to-offer conversion, offer-to-close conversion, average days per pipeline stage, and fee revenue per closed deal.

8. User Personas

Five personas cover the MVP; they are working archetypes drawn from your brief, not findings from user research, and should be revised after the interviews in section 3.

Persona	Type	Goal	Key concern	Main screens
Retiring owner (solo general dentist)	Primary, seller	Sell the practice and hand patients to a trusted successor	Staff and patients finding out early; fair price	Seller dashboard, sale request, documents, inquiries
Exiting or partner-selling owner (multi-chair or small group)	Primary, seller	Sell all or part; keep clinical role if possible	Partial-sale terms; confidentiality	Seller dashboard, transition type, offers
Associate-to-owner buyer	Primary, buyer	Buy a first practice	Verifying revenue; financing readiness	Buyer profile, listings, saved searches, interests
Established owner expanding	Secondary, buyer	Add a location or acquire a neighbour	Speed; clean due diligence	Matches, data room, offers
Brokerage manager and broker	Primary, internal	Run many deals without losing track	Stalled deals; missed follow-ups; document chaos	Broker CRM, pipeline, tasks, admin moderation

Secondary roles from the brief (investor or acquisition group, legal advisor, financial advisor) are invited to a single deal with limited access; they are not standalone MVP personas. The two administrator roles are covered in section 12.

9. User Journey Maps

The seller, buyer and broker journeys run in parallel through one deal, with the broker as the gatekeeper at every disclosure step. The flow diagrams are in the appendix (section 28).

Stage	Seller	Buyer	Broker
1. Enter	Registers and verifies email	Registers and verifies email	Created by an administrator, never self-registered
2. Onboard	Submits a sale request: category, city, transition type, confidentiality preference	Completes buyer profile: specialty, cities, budget, size	Receives the lead; assigned by the manager within the CRM
3. Prepare	Enters financials and uploads documents; receives a preliminary estimate	Uploads proof-of-funds and identity details for qualification	Reviews financials; verifies documents; schedules follow-up
4. List and discover	Previews anonymised listing; submits for review	Browses approved, anonymised listings; saves searches; expresses interest	Approves, edits or rejects the listing; shortlists buyers
5. NDA and access	Sees who is requesting access	Accepts NDA; requests confidential information	Approves each access request; sets scope and expiry
6. Diligence	Answers requests; replies to comments	Reviews documents in the data room; raises questions	Tracks outstanding items; reviews document status
7. Offer	Reviews offers (via broker)	Submits offer; counter-offers	Mediates; records every revision
8. Close	Signs agreements outside the platform in MVP	Same	Moves the deal through legal review and closing stages
9. Handover	Follows handover checklist	Follows handover checklist	Marks the transaction closed; records fee

Pain points each journey must remove: repeated emailing of the same documents (seller), unverifiable numbers (buyer), and deals dropping out of sight (broker).

10. Functional Requirements

Each module below lists a user story, requirements, business rules, permissions, data, errors and the release phase. All API errors share one shape, `{ success:false, error:{ code, message, details[] } }`, with 400 validation, 401 unauthenticated, 403 forbidden, 404 not found (also used when a user may not know a record exists), 409 conflict, 422 business-rule violation and 429 rate limit. Detailed acceptance tests are in section 24.

A. Authentication and onboarding (MVP)

Story. As a visitor I can register as a seller or buyer, verify my email and log in securely.

- Registration collects name, email, mobile, password (minimum 10 characters) and role (seller or buyer only).
- Email verification link is single-use and expires in 24 hours; password-reset link expires in 1 hour and is single-use.
- Sessions: 15-minute access token plus rotating refresh token in an httpOnly, Secure, SameSite cookie; refresh-token reuse revokes the whole token family.
- Profile holds professional details (dental registration number and state council, optional) and an optional organisation profile.
- Optional business verification in MVP is manual: the user uploads proof, an administrator approves, and the profile shows a verified badge.

Rules. Brokers and administrators are created by invitation only. Unverified-email users cannot request access to any listing. Verification badges are granted only by administrators.

Permissions and data. Any visitor may register; a user edits only their own profile. Data: `users`, `organizations`, `sessions`.

B. Seller portal (MVP)

Story. As a seller I can describe my practice, set confidentiality preferences and track my sale request.

- Sale request: practice category (dental only at launch), city and locality, exact address (private), chairs, practitioners, staff counts, years in operation, transition type (complete sale, partial sale, partnership transition), confidentiality preferences.
- Financials: annual revenue and expenses for the last three financial years, entered as numbers and supported by uploaded documents.
- Uploads: photographs and documents go to private storage; no photo is shown on any listing until a broker approves a version that hides identifying signage.

- Status path: Draft → Submitted → In review → Changes requested or Rejected or Approved → Listed → Under offer → Sold or Withdrawn.

Rules. A request starts as Draft and only the owner can edit it until submission; after submission only a broker can change listing fields, and the seller sees a change log. The seller can withdraw at any stage before Sold.

Permissions and errors. Seller reads and writes only their own practices; a submission with missing mandatory fields returns 422 listing each missing field.

C. Buyer portal (MVP)

Story. As a buyer I can state what I am looking for, browse approved listings and request more information.

- Buyer profile: buyer type (associate, owner expanding, group), specialty, preferred cities, budget range, preferred practice size, transition preferences, financing status.
- Browse, filter and sort; save favourites; express interest; request confidential information.
- Track every request in a single list with its status (Requested, NDA pending, Approved, Rejected, Expired).

Rules. A buyer must complete the profile and be qualified by a broker (a manual yes or no with a note) before a confidential-information request is routed to the seller's broker. Budget and proof of funds are visible to brokers only, never to other buyers.

D. Practice listings and discovery (MVP, extras in P2)

Story. As a buyer I can find relevant practices without ever seeing who the owner is until I am cleared.

Visibility is tiered and enforced on the server:

Tier	Who sees it	Fields
0 Teaser	Any visitor or logged-in buyer	Generic title (e.g. "General dental practice, South Bengaluru"), specialty, city and zone, revenue band, chairs, transition type
1 Verified buyer	Logged-in, email-verified buyers	Adds years in operation, practitioners, locations, equipment summary, staff counts (aggregated), verification status

Tier	Who sees it	Fields
2 NDA cleared	Buyers with accepted NDA and broker approval	Adds clinic name, exact address, detailed financials, asking price if the seller authorised it
3 Data room	Per document, per buyer	Individual documents by category

- Search by keyword, specialty, city, revenue band, chairs, transition type; sort by newest, revenue band, chairs; pagination at 20 per page.
- Revenue is shown as a band (e.g. ₹30–50 lakh) at Tiers 0–1; exact figures only at Tier 2 or above.
- P2: saved searches with alerts, recommended practices, featured listings (**needs legal review under the advertising rules in section 19** because paid placement could resemble advertising).

Rules. Only Approved listings appear; a withdrawn or sold listing disappears from search immediately. Anonymisation is enforced when building the response, never in the frontend.

E. Practice valuation (P2; MVP captures a valuation request as a CRM lead)

Story. As a seller I can get a preliminary range with visible assumptions, clearly labelled as not a certified valuation.

- Inputs: gross revenue, operating expenses, EBITDA, equipment value, assets, practitioners, patients (aggregate count only), years in operation, city, specialty, growth rate, owner dependency (low, medium, high).
- Methods offered: revenue multiple, earnings multiple on adjusted EBITDA or owner's earnings, and asset-based floor. Discounted cash flow is excluded from the MVP estimate because a small practice's cash flows depend heavily on the owner.
- Multiples and adjustments are configuration set by your brokerage and versioned; **no numbers ship until validated with real Indian transactions** (section 3).
- Output: a low–mid–high range, the assumptions used and the plain-text disclaimer "Preliminary estimate, not a certified valuation, financial advice or a guaranteed price."

Rules. Each estimate stores the configuration version and inputs so it is reproducible; a broker may override with a reasoned note.

F. Buyer–seller matching (P2; MVP uses manual broker shortlists)

Story. As a broker I get ranked buyer suggestions per listing and can override them.

- Specialty is a hard filter; everything else is scored. Default weights (configurable, to be confirmed by you): location 30, budget fit 25, revenue and size fit 20, transition type 15, other preferences 10.
- Each suggestion shows the score and the reasons per criterion; brokers can add or remove a buyer manually and the reason is logged.
- Matches are suggestions for brokers only; buyers and sellers never see scores.

G. Broker CRM (MVP, advanced analytics in P2)

Story. As a broker I can see every lead, practice, buyer and deal I own, and what I must do next.

- Records: leads, sellers, buyers, practices, each with owner, source and status. A lead is created by any seller sale request, buyer sign-up, valuation request or manual entry.
- Assignment: a brokerage manager assigns or reassigns leads; every change is logged.
- Follow-ups and tasks: due date, assignee, linked record, status; overdue tasks surface on the dashboard and by email.
- Communication history: calls, meetings and messages logged manually in MVP; email sync is P2.
- Internal notes are visible to brokers and managers only and are never returned by any seller or buyer endpoint.
- Kanban pipeline with configurable stages (default stages in section K), drag to move, with an optional required-fields rule per stage.
- Analytics MVP: deals per stage, deals per broker, average days in stage. P2: conversion funnels, fee forecast.

Rules. A broker sees only deals assigned to them; a manager sees all deals in the brokerage. Moving a deal backwards requires a reason.

H. Confidentiality and NDA (MVP basic)

Story. As a seller I decide who learns my identity, and every disclosure is recorded.

- NDA templates are created by an administrator, versioned, and marked approved only after your legal advisor signs them off.
- A buyer accepts the current NDA version with a clickwrap step. The system stores the user, NDA version and document hash, timestamp and IP address.
- **Legal position.** The MVP records acceptance; it does not claim to create a legally enforceable electronic signature. Whether clickwrap is sufficient, or whether an e-sign provider is needed, is a question for counsel before launch (P3 integration).

- Access states: Not requested → Requested → NDA sent → NDA accepted → Broker approved or Rejected → Expired or Revoked.
- An access grant has a scope (tier, document categories), an expiry date and a revocation reason; revoking takes effect on the next request.
- Every view or download of confidential data writes an access-log entry.

Rules. Tier 2 data is released only when the NDA is accepted and a broker has approved; the seller can veto any individual buyer.

I. Document management and due diligence (MVP uploads, P2 workflows)

Story. As a seller or broker I can upload, categorise and share documents safely.

- Categories: financial statements, tax records, licences and registrations, equipment inventory, property lease, employment agreements, insurance, operational reports, acquisition agreements.
- Upload checks: allowed types PDF, DOCX, XLSX, JPG, PNG; maximum 25 MB per file (**proposed; confirm**); content-type sniffing, not extension only; malware scan before a file becomes available.
- Storage is private object storage; files are served only through short-lived (60-second) signed URLs after a server-side permission check.
- Versioning: re-uploads create a new version and keep the old; expiry dates for time-bound documents; review status Pending, Verified, Rejected, Needs update, set by a broker.
- P2: comments per document, approval workflows, checklist templates per deal, document-completeness hints.

J. Offers and negotiation (P2)

Story. As a broker I can record, track and compare offers and counter-offers.

- Fields: proposed price, deal structure (asset sale, share sale, partnership buy-in), conditions, expiry date, notes.
- Every revision is a new immutable record linked to its parent; withdrawn and expired offers stay in the history.
- Offers pass through the broker; buyer and seller never see each other's identity until Tier 2.
- Accept or reject by the seller records a decision and moves the deal to the next stage.

K. Transaction management (MVP pipeline, P2 requirement checklists)

Story. As a broker I can see where each deal stands and what is still outstanding.

Default pipeline stages (configurable by managers): New lead → Buyer qualified → NDA → Information disclosure → Due diligence → Initial offer → Negotiation → Letter of intent → Legal review → Final agreement → Closing → Ownership transition → Post-sale handover, plus Lost and Withdrawn.

- Each stage carries a checklist of requirements; the deal cannot enter a stage marked "hard gate" until the checklist is complete or a manager overrides with a reason.
- Legal review, final agreement and closing happen outside the platform in MVP; the system records dates, parties and uploaded executed copies.

L. Communication and notifications (MVP in-app and email; P2 messaging)

- MVP: in-app notifications and email for access requests, NDA status, document review, deal stage change and follow-up reminders; users can mute non-critical types.
- P2: broker-mediated messaging inside a deal; direct buyer–seller messaging stays off unless a manager enables it for a deal.
- Emails never contain confidential figures or document names; they link to the secure app.

M. Admin dashboard (MVP)

- Counts: registered users, active buyers, active sellers, verified practices, pending listings, active deals, closed transactions; revenue metrics from recorded fees (P2).
- Functions: user management (invite brokers, suspend users), listing moderation queue, verification queue, NDA template management, and a searchable platform activity log.
- Moderation outcomes: Approve, Request changes, Reject, each with a reason shown to the seller.

11. Non-Functional Requirements

The targets below are proposals sized for an MVP with tens of brokers and a few thousand users; change them if your expected load differs.

Area	Requirement (proposed)	How it is checked
Performance	Listing search returns in under 800 ms at the 95th percentile with 5,000 listings; API reads under 500 ms at p95	Load test before launch

Area	Requirement (proposed)	How it is checked
Availability	99.5% monthly uptime for the MVP; planned maintenance outside 9:00–20:00 IST	Uptime monitor
Data residency	Primary data and file storage hosted in an Indian cloud region (**confirm with counsel**)	Deployment review
Backup and recovery	Daily database snapshots kept 30 days; point-in-time recovery if the database tier supports it; recovery point 24 hours, recovery time 4 hours	Quarterly restore drill
Security	OWASP ASVS Level 2 as the target baseline; full detail in section 19	Security tests, section 23
Privacy	Consent, notice and breach handling designed against the DPDP Act and Rules (section 19)	Counsel review
Accessibility	WCAG 2.1 AA for all authenticated and public pages	Automated and manual audit
Browser support	Last two versions of Chrome, Edge, Safari and Firefox; Android Chrome and iOS Safari	Cross-browser test run
Responsiveness	Fully usable from 360 px width; broker CRM optimised for 1280 px and up	Responsive tests
Scalability	Stateless API behind a load balancer; file storage and database scale independently	Architecture review
Observability	Structured logs with request IDs; error tracking; audit log separate from application log	Runbook
Maintainability	TypeScript strict mode; 80% unit-test coverage on services and policies; lint and type checks block merges	CI
Language	English at launch; copy kept in a message catalogue so Hindi and other languages can be added	Code review

12. Role and Permission Matrix

Permissions are enforced on the Express server by a policy function that combines the user's role with ownership, assignment and deal-access checks; hiding a button in the frontend is never the control. Administrators have no standing access to confidential documents, only a logged break-glass path, which keeps the least-privilege promise sellers will ask about.

Codes: All = every record · **Own** = records the user created · **Asg** = records assigned to them · **Grant** = only what a deal-scoped access grant allows · **–** = no access.

Roles: SA Super Admin · PA Platform Admin · BM Brokerage Manager · BR Broker · SE Seller · BU Buyer · IN Investor or acquisition group · LA Legal Advisor · FA Financial Advisor. IN is a buyer subtype with an organisation. LA and FA are invited to one deal at a time (P2).

Capability	SA	PA	BM	BR	SE	BU	IN	LA	FA
Invite or suspend users	All	All except SA	–	–	–	–	–	–	–
Manage NDA templates	All	All	View	View	–	–	–	–	–
Create or edit a listing	–	–	All	Asg	Own, draft only	–	–	–	–
Approve a listing	All	All	All	– (can recommend)	–	–	–	–	–
View Tier 0 and Tier 1 listing	All	All	All	All	Own	All verified	All verified	Grant	Grant
View Tier 2 (identity, financials)	Break-glass, logged	Listing fields only	All	Asg	Own	Grant	Grant	Grant	Grant
Upload documents	–	–	All	Asg	Own	Own qualification documents	Own qualification documents	Grant	Grant
Verify documents	All	–	All	Asg	–	–	–	–	–
Download data-room documents	Break-glass, logged	–	All	Asg	Own	Grant	Grant	Grant	Grant
Approve an access request	–	–	All	Asg	Veto only	–	–	–	–
Create or revise an offer (P2)	–	–	Record on behalf	Record on behalf	Counter-offer	Own	Own	–	–

Capability	SA	PA	BM	BR	SE	BU	IN	LA	FA
Withdraw an offer (P2)	–	–	–	Record on behalf with reason	–	Own	Own	–	–
Accept or reject an offer (P2)	–	–	–	Record only	Own	–	–	–	–
Move deals in the pipeline	View	View	All	Asg	–	–	–	–	–
Read internal notes	Break-glass, logged	–	All	Asg	–	–	–	–	–
View audit logs	All	All	Own brokerage	–	–	–	–	–	–
View analytics	All	All	Own brokerage	Own	–	–	–	–	–
Configure pipeline stages	All	–	All	–	–	–	–	–	–

Dashboards by role

Role	Dashboard
SA, PA	Admin: user counts, moderation and verification queues, platform activity
BM	Brokerage: all deals, workload per broker, overdue tasks, listings pending approval
BR	My work: assigned leads, today's tasks, my pipeline
SE	My practice: sale status, document checklist, access requests, notifications
BU, IN	My search: matches, saved searches, interests, access status
LA, FA	Deal-scoped view of granted documents and comments only (P2)

Who does what, in one line each

- **Approves listings:** a brokerage manager or administrator; brokers recommend, sellers cannot self-approve.
- **Verifies documents:** the assigned broker or a manager.

- **Accesses confidential financials:** the seller, the assigned broker, managers, and buyers or advisors with an approved, unexpired grant.
- **Creates, edits or withdraws offers:** the buyer owns the offer; brokers record changes on the buyer's written instruction and must add a reason when withdrawing.
- **Manages transactions:** the assigned broker and the brokerage manager.

13. Complete Feature Inventory

The MVP has 35 features, 31 of them Must-haves; Phase 2 and 3 features are listed so dependencies are visible. Value and complexity are my estimates (H, M, L) and should be recalibrated by your engineers. Acceptance criteria for the critical features are in section 24 and referenced by ID.

ID	Feature	Phase	MoSCoW	Value	Complexity	Depends on
F01	Registration and email verification	MVP	Must	H	L	–
F02	Login, secure sessions, password reset	MVP	Must	H	M	F01
F03	Server-side RBAC and policy checks	MVP	Must	H	M	F02
F04	Role-based onboarding and profiles	MVP	Must	H	L	F02
F05	Manual business and registration verification	MVP	Should	M	L	F04
F06	Practice sale request (draft, submit)	MVP	Must	H	M	F04
F07	Three-year financial intake	MVP	Must	H	M	F06
F08	Confidentiality preferences	MVP	Must	H	L	F06
F09	Buyer profile and preferences	MVP	Must	H	L	F04
F10	Broker qualification of buyers	MVP	Must	H	L	F09
F11	Listing moderation and approval	MVP	Must	H	M	F06
F12	Tiered anonymised listing responses	MVP	Must	H	H	F03, F11
F13	Search, filters, sort, pagination	MVP	Must	H	M	F12
F14	Express interest; favourites	MVP	Must (interest), Should (favourites)	M	L	F12

ID	Feature	Phase	MoSCoW	Value	Complexity	Depends on
F17	NDA templates and clickwrap acceptance	MVP	Must	H	M	F03
F18	Access requests, approval, expiry, revocation	MVP	Must	H	H	F17
F19	Access and download logging	MVP	Must	H	M	F18
F20	Private upload with type, size and malware checks	MVP	Must	H	H	F03
F21	Document categories, versions, expiry, review status	MVP	Must	H	M	F20
F22	Signed-URL downloads after permission check	MVP	Must	H	M	F20, F18
F25	Lead, seller, buyer, practice records	MVP	Must	H	M	F03
F26	Lead assignment	MVP	Must	H	L	F25
F27	Tasks, follow-ups, reminders	MVP	Must	H	M	F25
F28	Communication log and internal notes	MVP	Must	H	L	F25
F29	Kanban pipeline, configurable stages	MVP	Must	H	M	F25
F30	Basic broker and admin analytics	MVP	Should	M	M	F29
F32	Valuation request captured as lead	MVP	Should	M	L	F25
F34	Manual broker buyer shortlist	MVP	Must	H	L	F10, F29
F37	In-app and email notifications	MVP	Must	H	M	F03
F39	Admin dashboard KPIs	MVP	Must	M	M	F25
F40	User management and broker invitations	MVP	Must	H	L	F03
F41	Audit log viewer	MVP	Must	H	M	F19
F42	Rate limiting, security headers, input validation	MVP	Must	H	M	–
F43	Backups, monitoring, error tracking	MVP	Must	H	M	–
F36	Basic offer record in the deal (notes and amount)	MVP	Should	M	L	F29
F15	Saved searches and alerts	P2	Should	M	M	F13, F37
F16	Featured and recommended listings	P2	Could	L	M	F13 (legal review first)

ID	Feature	Phase	MoSCoW	Value	Complexity	Depends on
F23	Document comments and approval workflow	P2	Should	M	M	F21
F24	Due-diligence checklists, completeness hints	P2	Should	H	M	F21
F31	Advanced analytics and funnels	P2	Could	M	M	F30
F33	Preliminary valuation estimator	P2	Should	H	M	F07, validated data
F35	Automated matching score	P2	Should	H	M	F09, F12
F38	Broker-mediated deal messaging	P2	Should	M	M	F37
F49	Invited legal and financial advisor access	P2	Should	M	M	F18
F51	Full offer and counter-offer management	P2	Should	H	M	F36
F44	E-signature integration	P3	Could	H	H	F17
F45	Financing and lender integration	P3	Could	M	H	F09
F46	AI-assisted features (section 20)	P2–P3	Could	M	M–H	F21
F47	External brokers and multi-organisation support	P3	Could	M	H	F03
F48	Advanced reporting	P3	Could	M	M	F31

Won't have (this roadmap): patient-identifiable records, payment collection, for-sale-by-owner public listings, native mobile apps, automated certified valuations.

14. Screen Inventory

The MVP needs 40 screens. To avoid repeating the same text 40 times, loading, error, success and responsive behaviour are defined once as global patterns below; each screen row then adds only what is specific to it (purpose, components, actions, empty state).

Global state patterns (apply to every screen unless a row overrides them)

State	Behaviour
Loading	Skeleton placeholders matching the final layout; no spinner-only pages; buttons show an inline spinner and are disabled while saving

State	Behaviour
Error	Inline error banner with plain-language message and a Retry action; field errors appear under the field with `aria-describedby`; 401 redirects to login, 403 shows a no-access page with a way back
Success	A toast for 4 seconds for quick actions (saved, sent); a confirmation page for irreversible actions (submitted, accepted NDA)
Responsive	Desktop (1280 px and up): sidebar plus content. Tablet (768–1279 px): collapsible sidebar, two-column grids become one. Mobile (below 768 px): bottom navigation for the three most-used actions, tables become stacked cards, Kanban becomes a stage selector with a card list

Public and authentication

ID	Screen	Role	Purpose and main components	Key actions	Empty state
S01	Landing	Visitor	Value proposition, how it works, tier-0 teaser listings, sign-up call to action	Sign up as seller or buyer	No teasers: show how-it-works only
S02	Register	Visitor	Form: name, email, mobile, password, role	Create account	–
S03	Login	Visitor	Email and password; lockout message after repeated failures	Log in, reset password	–
S04	Verify email	Visitor	Confirmation of link outcome	Resend link	–
S05	Forgot and reset password	Visitor	Request link; set new password	Reset	–

Onboarding and profile

ID	Screen	Role	Purpose and main components	Key actions	Empty state
S06	Seller onboarding wizard	Seller	Three steps: about you, about the practice, confidentiality	Save draft, continue	–
S07	Buyer onboarding wizard	Buyer, Investor	Three steps: buyer type, preferences, budget and financing	Save, continue	–

ID	Screen	Role	Purpose and main components	Key actions	Empty state
S08	Profile and organisation	All external	Personal and professional details, organisation	Edit, save	–
S09	Verification submission	Seller, Buyer	Upload proof; status tracker	Submit, resubmit	No submission yet: explain benefit of verification

Seller

ID	Screen	Role	Purpose and main components	Key actions	Empty state
S10	Seller dashboard	Seller	Sale status timeline, document checklist progress, pending access requests, notifications	Open request, upload	"Start your sale request" prompt
S11	Sale request form	Seller	Multi-step form: practice details, transition type, address (private), staff counts	Save draft, submit for review	–
S12	Financial intake	Seller	Three-year revenue and expense table with document attachments	Save, attach	Empty rows with examples
S13	Documents	Seller	Checklist by category, versions, expiry, review status	Upload, replace, delete draft files	Checklist with all categories open
S14	Listing preview	Seller	Exactly what each tier will see	Back, submit	–
S15	Access requests	Seller	Buyers (anonymised until cleared), status, broker note	Veto, view history	"No requests yet"

Buyer

ID	Screen	Role	Purpose and main components	Key actions	Empty state
S17	Buyer dashboard	Buyer	Profile completeness, recent listings, request statuses	Open listing	"Complete your profile to unlock requests"
S18	Listing search	Buyer	Filters, sort, result cards, pagination	Filter, save favourite, open	"No practices match: widen filters" with a clear-filters action
S19	Listing detail	Buyer	Tiered fields, locked-field placeholders, interest and access request buttons	Express interest, request information	–

ID	Screen	Role	Purpose and main components	Key actions	Empty state
S20	Favourites	Buyer	Saved listings	Remove, open	"Save listings to compare later"
S21	My requests	Buyer	Interests and access requests with status	Cancel, open	"No requests yet"
S22	NDA acceptance	Buyer	NDA text, version, checkbox, typed full name	Accept, decline	–

Broker

ID	Screen	Role	Purpose and main components	Key actions	Empty state
S23	Broker dashboard	Broker, Manager	Today's tasks, overdue follow-ups, my deals by stage, new leads	Open task, open lead	"No tasks today"
S24	Leads	Broker, Manager	Filterable table: source, status, owner, last contact	Assign, create lead, bulk assign	"No leads yet"
S25	Seller, buyer or lead detail	Broker, Manager	Profile, activity timeline, notes, tasks, linked practice	Add note, add task, qualify	–
S26	Practice detail	Broker, Manager	All practice fields (Tier 2), documents, listing status	Edit, submit for approval	–
S27	Pipeline	Broker, Manager	Kanban of deals by stage, filters by broker	Drag card, open deal	Empty column hints
S28	Deal workspace	Broker, Manager	Stage checklist, parties, offers (basic), documents, activity	Advance stage, record offer	–
S29	Data room	Broker, Manager, granted external	Document tree by category, review status, access log	Verify, grant, revoke	"No documents uploaded"
S30	Tasks and calendar	Broker	List and calendar of follow-ups	Complete, reschedule	"Nothing due"
S31	Buyer shortlist	Broker	Qualified buyers for a listing, manual add or remove	Add, remove, share	"No shortlisted buyers"
S32	Analytics	Broker, Manager	Deals per stage, days per stage, per-broker counts	Change date range	"Not enough data yet"

Administration and shared

ID	Screen	Role	Purpose and main components	Key actions	Empty state
S33	Admin dashboard	SA, PA	KPI tiles, queues	Open queue	–
S34	Listing moderation	SA, PA, Manager	Queue with diff view and reasons	Approve, request changes, reject	"Queue clear"
S35	Verification queue	SA, PA	Submitted proofs	Approve, reject	"Queue clear"
S36	User management	SA, PA	Search, roles, suspend, invite brokers	Invite, suspend, reset	–
S37	NDA templates	SA, PA	Versions, approval flag, preview	Create version, mark approved	"No template yet"
S38	Pipeline configuration	SA, Manager	Stage names, order, gates	Add, reorder, retire stage	–
S39	Audit log	SA, PA, Manager	Searchable event list with filters	Filter, export	"No events"
S40	Not found, no access, maintenance	All	Clear message and a way back	Go back	–

Notifications (S16) is a shared drawer, not a page: unread count, grouped by day, mark as read, deep links into the item.

15. UI/UX Guidelines

The visual direction is calm, precise and confidential-by-default: a quiet B2B workspace closer to a deal-room tool than a consumer marketplace. I could not inspect competitor screenshots or logged-in screens, so I derived patterns only from public page structure: coded listing cards with a few key metrics (Henry Schein DPT), area-level masking of location (Dental Acquisitions), and stepwise seller processes (National Practice Sales). Treat the visual choices below as my proposal, to be validated with a short usability test.

Brand positioning. Discreet, expert, fair. Copy is plain, never promotional; every screen that shows sensitive data says who can see it.

Design principles

- Show who can see this: a lock icon and "Visible to: you and your broker" label on every field group that is restricted.
- Locked fields in buyer views show a placeholder ("Available after NDA"), never blank space or a hidden element, so buyers understand what they will get.
- Sellers get spacious, guided screens; brokers get dense, scannable tables and keyboard shortcuts.

- Numbers are never decorative: no gradient charts or vanity widgets.

Colour tokens (WCAG contrast checked against white or the surface colour)

Token	Hex	Use
Primary	#0F4C5C	Buttons, links, active navigation
Primary hover	#0B3A47	Hover and pressed
Ink	#1B2430	Body and headings
Muted text	#5B6573	Secondary text (5.9:1 on white)
Surface	#F7F8FA	Page background
Border	#E1E5EB	Dividers and inputs
Success	#1F7A4D	Verified, approved
Warning text	#8A5A00	Pending, expiring (never use a lighter amber for text)
Danger	#B42318	Errors, revoked
Info	#2457A6	Notices, links in notes

Status is always shown as an icon plus a word, never colour alone.

Typography. Inter for all interface text, self-hosted through Next.js font loading. Scale: 12, 14 (default), 16, 18, 20, 24, 30, 36 px, with 1.5 line height for body and 1.25 for headings. Noto Sans Devanagari is reserved for the later Hindi release.

Spacing and shape. 4 px base unit; common steps 8, 12, 16, 24, 32, 48. Radius 8 px on cards, 6 px on inputs and buttons. One subtle elevation level only, used for drawers and menus.

Component system. Build once, reuse everywhere, on Tailwind with accessible headless primitives (Radix UI, the layer under shadcn/ui, justified because dialogs, menus and comboboxes are error-prone to build accessibly).

Group	Components
Inputs	Text, number with ₹ formatting, select, combobox, date, multi-step form, file uploader with progress and validation
Data display	Table with sort and filter, status pill, Kanban card, timeline, key-value list, locked-field placeholder

Group	Components
Feedback	Toast, banner, skeleton, empty state, confirm dialog
Navigation	Sidebar, top bar, tabs, breadcrumb, stepper, pagination
Overlays	Modal, drawer (notifications, filters on mobile), tooltip

Layouts and navigation. A persistent left sidebar by role (collapsed to icons on tablet), a top bar with global search, notifications and the profile menu, and breadcrumbs inside any deal. Mobile uses a bottom bar with three role-specific shortcuts. Dashboards use a 12-column grid with KPI tiles on top and work queues below.

Accessibility. WCAG 2.1 AA: full keyboard operation, visible 2 px focus ring in Primary, labels bound to inputs, error summaries focused on submit, 44 px minimum touch targets on mobile, and no information conveyed by colour alone.

16. Database Schema

The MVP needs 21 MongoDB collections, designed so that the anonymised listing is its own document and can never leak private practice fields. Mongoose schemas enforce types and required fields (* below); the API validates with Zod before anything reaches Mongoose. All documents carry `createdAt`, `updatedAt` and an optimistic-concurrency version (`__v`); records that users may delete are soft-deleted with `deletedAt`.

Core collections

Collection	Purpose	Key fields (* required)	Indexes	Access
`users`	Accounts and roles	email*, passwordHash*, role*, status*, name*, mobile, orgId, professional{regNo, council}, emailVerifiedAt, failedLogins, lockedUntil	unique email; role+status	Owner reads own; admins read basic fields
`organizations`	Practices, buyer groups, brokerage	name*, type*, gstin, address, verifiedAt, verifiedBy	name; type	Members; admins
`buyerProfiles`	Buyer preferences and qualification	userId* (unique), buyerType*, specialties[], cities[], budgetMin, budgetMax, sizePreference, transitionTypes[], financingStatus, qualification{status, by, at, note}	cities; specialties; qualification.status	Owner; brokers; **budget hidden from other buyers**
`sellerProfiles`	Seller context	userId* (unique), exitTimeline, defaultConfidentiality	userId	Owner; brokers

Collection	Purpose	Key fields (* required)	Indexes	Access
`practices`	Full private practice record	ownerId*, brokerId, specialty*, name*, address{line, city*, zone, state, pin}, yearsInOperation, practitioners, chairs, locations, staffCounts, equipmentSummary, transitionType*, confidentiality{showAskingPrice}, status*	ownerId; brokerId+status; address.city+specialty+status	Owner; assigned broker; managers
`practiceFinancials`	Per-year confidential financials	practiceId*, fiscalYear*, revenue*, expenses*, ebitda, askingPrice, source	unique practiceId+fiscalYear	Tier 2 grants; owner; broker
`listings`	Anonymised, approved view	practiceId* (unique), publicTitle*, teaser{specialty, city, zone, revenueBand, chairs, transitionType}, tier1{years, practitioners, locations, equipmentSummary, staffAggregate}, approvedBy, approvedAt, status*, publishedAt	status+teaser.city+teaser.specialty+publishedAt; text index on publicTitle	Any verified user (tiers 0–1 only)
`leads`	CRM entry point	type*, source*, contact{name, email, phone}, userId, assignedTo, status*, nextFollowUpAt	assignedTo+status; nextFollowUpAt	Brokers; managers
`buyerInterests`	Interests and information requests	buyerId*, listingId*, kind* (interest or info_request), status*, brokerNote	unique buyerId+listingId+kind	Buyer own; assigned broker
`ndaTemplates`	Versioned NDA text	version*, title*, bodyHash*, storageKey*, approvedByLegalAt, active	unique version	Admins write; brokers read
`ndaAcceptances`	Proof of acceptance	userId*, templateId*, version*, bodyHash*, typedName*, acceptedAt*, ip*, userAgent	userId+templateId	Admins; assigned broker
`accessGrants`	Who may see what	buyerId*, listingId or dealId*, tier*, categories[], status*, grantedBy*, expiresAt*, revokedAt, revokedReason	buyerId+listingId; expiresAt	Brokers; the buyer reads own
`documents`	Document metadata (never the file)	practiceId*, dealId, category*, title*, current{v, storageKey, mime, sizeBytes, sha256}, versions[], scanStatus*, reviewStatus*, expiresOn	practiceId+category; reviewStatus	Owner; brokers; grants
`deals`	Transaction records	practiceId*, sellerId*, buyerId, brokerId*, stageId*, stageHistory[], status*, offerSummary{amount, notes}, fee{amount, status}	brokerId+stageId; practiceId; status	Assigned broker; managers; seller sees own stage only
`pipelineStages`	Configurable stages	name*, order*, gate*, checklist[], active	order	Managers write; brokers read

Collection	Purpose	Key fields (* required)	Indexes	Access
`tasks`	Follow-ups	title*, assigneeId*, dueAt*, status*, leadId, dealId	assigneeId+status+dueAt	Assignee; managers
`activities`	Notes, calls, system events	entityType*, entityId*, kind*, body, authorId*, visibility* (always internal for notes)	entityType+entityId+createdAt	Brokers; managers
`notifications`	In-app notifications	userId*, type*, payload, readAt	userId+readAt+createdAt; TTL 180 days	Owner
`auditLogs`	Append-only audit trail	actorId*, actorRole*, action*, entityType*, entityId*, ip, meta, at*	at; actorId+at; entityType+entityId	Admins; managers (own brokerage); insert-only database user
`refreshTokens`	Rotating sessions	userId*, familyId*, tokenHash*, expiresAt*, revokedAt, replacedBy	TTL on expiresAt; familyId	Server only
`verificationRequests`	Optional verification	subjectId*, kind*, documents[], status*, reviewedBy, reviewedAt	status+createdAt	Admins; subject reads own

Phase 2 collections: `offers` (parent link, immutable revisions), `matches` (score and per-criterion reasons), `ddChecklists`, `messages`, `savedSearches`, `valuationConfigs` (versioned multiples), `valuationEstimates` (inputs and config version).

Embedding versus referencing

- **Embed** bounded, always-read-together data: document versions in `documents`, stage history in `deals` (about 15 entries), the anonymised tiers inside `listings`.
- **Reference** anything that grows or has its own access rule: financials, activities, tasks, audit logs, grants.
- **Denormalise on purpose** only the anonymised listing, generated at approval; changing a practice's city or revenue later does not alter the live listing until a broker re-approves it.

Sensitive-data handling. Financial figures, exact address and the practice name are encrypted at the application level (AES-256-GCM with a key held in a cloud key-management service, not in the database or repo), in addition to encryption at rest. Passwords use bcrypt (cost 12). File bytes never live in MongoDB; `storageKey` points to private object storage.

Example: practice, financial year and listing (abridged)

```
{
  "practice": {
    "_id": "pr_01", "ownerId": "usr_88", "brokerId": "usr_12",
    "specialty": "general", "name": "<encrypted>",
    "address": { "line": "<encrypted>", "city": "Bengaluru", "zone": "South",
"state": "KA", "pin": "<encrypted>" },
    "yearsInOperation": 14, "practitioners": 2, "chairs": 4,
    "transitionType": "complete_sale", "status": "listed"
  },
  "financial": { "practiceId": "pr_01", "fiscalYear": "2025-26", "revenue": "
<encrypted>", "expenses": "<encrypted>" },
  "listing": {
    "practiceId": "pr_01", "publicTitle": "General dental practice, South
Bengaluru",
    "teaser": { "specialty": "general", "city": "Bengaluru", "zone": "South",
"revenueBand": "50L-1Cr", "chairs": 4, "transitionType": "complete_sale" },
    "status": "approved", "approvedBy": "usr_03", "approvedAt": "2026-11-
01T09:30:00Z"
  }
}
```

Example: access grant

```
{
  "buyerId": "usr_41", "listingId": "lst_07", "tier": 2,
  "categories": ["financial_statements", "licences"],
  "status": "active", "grantedBy": "usr_12",
  "expiresAt": "2026-12-31T23:59:59Z", "revokedAt": null
}
```

The collection relationship diagram is in the appendix (section 28).

17. API Specification

The Express API is versioned under `/api/v1`, speaks JSON, and checks authorisation on every route; this section fixes the conventions and specifies the 13 endpoints that carry the confidentiality rules. The full MVP inventory (97 endpoints) is in the appendix (section 28).

Conventions

Topic	Rule
Authentication	Short-lived access token in the `Authorization: Bearer` header; rotating refresh token in an httpOnly, Secure, SameSite=Strict cookie, refreshed through `POST /auth/refresh` with a custom CSRF header
Success envelope	`{ "success": true, "data": …, "meta": { … } }`
Error envelope	`{ "success": false, "error": { "code": "VALIDATION_FAILED", "message": "…", "details": [{ "field": "email", "issue": "…" }] } }`
Pagination	`page` (default 1) and `pageSize` (default 20, maximum 100); `meta` returns `page, pageSize, total, totalPages`
Sorting	`sort=field:asc` or `field:desc`, restricted to a whitelist per endpoint; unknown fields return 400
Filtering	Plain query parameters per endpoint (e.g. `city`, `specialty`, `revenueBand`); repeated values mean OR
Validation	Zod schema per route; unknown body fields are rejected, not silently stored
Idempotency	`Idempotency-Key` header on POSTs that create interests, access requests and offers
Rate limits (proposed)	Auth endpoints 10 per minute per IP; login lockout after 5 failures in 15 minutes per account and IP; general 300 per minute per user; uploads 20 per hour per user; access requests 20 per day per buyer. Exceeding returns 429 with `Retry-After`
Route modules	`auth`, `users`, `profiles`, `practices`, `listings`, `interests`, `nda`, `access`, `documents`, `leads`, `deals`, `pipeline`, `tasks`, `activities`, `notifications`, `admin`, `audit`

Critical endpoints

Endpoint	Roles	Request	Success	Errors
`POST /auth/register`	Public	`{ name, email, mobile, password, role: "seller"|"buyer" }`; password at least 10 characters	201 `{ userId }`; verification email sent	409 email exists; 422 weak password; 400 role other than seller or buyer
`POST /auth/login`	Public	`{ email, password }`	200 `{ accessToken, user }` plus refresh cookie	401 invalid credentials (same message for unknown email); 423 account locked; 403 email not verified
`POST /auth/refresh`	Cookie	Refresh cookie plus CSRF header	200 new access token and rotated cookie	401 invalid or reused token (the whole token family is revoked)
`POST /practices`	Seller, Broker	Core fields from section 10B; saved as `draft`	201 `{ practice }`	400 unknown field; 422 invalid enum (e.g. transition type)
`POST /practices/:id/submit`	Seller (owner)	Empty body	200 status `submitted`	422 missing mandatory fields (each listed); 403 not owner; 409 already submitted
`GET /listings`	Buyer, Investor, Broker, Admin	Query: `q, city, specialty, revenueBand, chairs, transitionType, sort, page, pageSize`	200 list of Tier 0–1 cards only	400 bad sort field; 401; 403 if email not verified
`GET /listings/:id`	Buyer, Investor, Broker, Admin	–	200 fields allowed by the caller's tier, with `lockedFields[]` naming what is withheld	404 if not approved or sold (never reveals existence of drafts)
`POST /listings/:id/interests`	Verified buyer	`{ kind: "interest"|"info_request", message? }`	201 `{ interestId, status }`	409 duplicate; 422 buyer profile incomplete or not yet qualified (for `info_request`)
`POST /nda/accept`	Buyer	`{ templateVersion, listingId, typedName }`	201 `{ acceptanceId }`; stores hash, time and IP	409 template version outdated; 422 typed name does not match profile name
`POST /access-requests/:id/approve`	Broker (assigned), Manager	`{ tier, categories[], expiresAt }`	200 grant created	403 not assigned; 422 NDA not accepted or seller veto recorded; 400 expiry in the past
`POST /documents/uploads`	Seller, Broker, Buyer (own qualification)	`{ practiceId, category, fileName, mime, sizeBytes }`	201 `{ documentId, uploadUrl, expiresIn }` (pre-signed, single object, 10 minutes)	400 type not allowed; 413 over 25 MB; 429 too many uploads

Endpoint	Roles	Request	Success	Errors
`GET /documents/:id/download-url`	Owner, Broker, Grant holder	–	200 `{ url, expiresIn: 60 }`; writes an access-log entry	403 no active grant; 404 unknown or not shared with the caller; 409 file not yet scanned clean
`PATCH /deals/:id/stage`	Broker (assigned), Manager	`{ stageId, reason? }`	200 deal with updated `stageHistory`	422 hard-gate checklist incomplete (list returned); 422 backward move without reason; 403 not assigned

Authorisation order inside every handler: authenticate, authorise by role, check ownership or assignment or grant, validate input, run the service, write the audit entry, return. A request that fails an earlier step never reaches a later one, and 404 is used instead of 403 wherever a 403 would reveal that a confidential record exists.

18. System Architecture

The system is a Next.js frontend talking to a separate Express API over HTTPS, with MongoDB for data and private object storage for files, exactly as your mandated stack requires; Next.js API routes are not used. The Mermaid architecture diagram is in the appendix (section 28).

Components

Component	Responsibility	Technology
Web app	Pages, forms, role-based dashboards; no business rules and no secrets	Next.js, React, TypeScript, Tailwind CSS
API	Authentication, authorisation, validation, business rules, audit logging	Node.js, Express.js, TypeScript
Database	All structured data	MongoDB with Mongoose
Object storage	All uploaded files, private bucket, served only by short-lived signed URLs	S3-compatible storage in an Indian region
Job worker	Malware scanning, email sending, reminder scheduling	BullMQ on Redis, same codebase as the API
Email	Transactional email	A transactional email provider (to be selected; SES is one option)

Component	Responsibility	Technology
Observability	Logs, error tracking, uptime	Pino structured logs, Sentry, an uptime monitor

Recommended supporting tools and why

Tool	Why it earns its place
Zod (shared package)	One schema validates the form in the browser and the request in the API, so rules cannot drift apart
React Hook Form	Long multi-step forms (sale request, financial intake) with minimal re-rendering
TanStack Query	Caching, retries and loading and error states for server data
bcrypt (cost 12)	Password hashing as specified; argon2 is stronger but not required
Helmet, express-rate-limit with a Redis store, cors	Security headers, limits that survive multiple API instances, strict origin list
Private object storage instead of Cloudinary	Confidential documents need private buckets, signed URLs and access logs; Cloudinary suits public media, so use it only if you also want image transformation for photos
BullMQ and Redis	Scans and emails must not block requests; justified only because uploads are scanned
ClamAV	Open-source malware scanning of uploads before they become available
Vitest, Supertest, Playwright, MongoDB Memory Server	Unit and component tests, HTTP tests, browser tests, and fast database tests

How a confidential download flows

1. The browser calls the API with the access token.
2. The API authenticates, checks role, ownership or assignment, and an active unexpired grant.
3. The API writes an audit entry, then asks storage for a 60-second signed URL.

4. The browser downloads directly from storage; the URL cannot be reused after it expires.

Deployment (to be confirmed). I recommend containers for web and API behind a load balancer, MongoDB on a managed replica set (replica sets are needed for Mongoose transactions), and separate staging and production environments. Hosting provider and region are an open question (section 27) because they drive data-residency advice.

Recommended repository structure. I changed your example in one way: a shared package for Zod schemas and types, so frontend and backend validate identically.

```
project-root/
  apps/
    web/                 # Next.js
      app/ components/ features/ hooks/ services/ lib/ types/ public/
    api/                 # Express
      src/
        config/ controllers/ services/ models/ routes/
        middleware/ policies/ validators/ jobs/ utils/ types/
      tests/
  packages/
    shared/              # Zod schemas, enums, types used by both apps
  docs/
    PRD.md  API.md  DATABASE.md  ARCHITECTURE.md  DECISIONS.md
  .github/workflows/     # CI: lint, type-check, test, build
  pnpm-workspace.yaml
  README.md
```

The `policies/` folder holds one function per protected action (for example `canDownloadDocument(user, document, grant)`), so every authorisation rule is in one place and unit-testable.

19. Security Architecture

The platform will hold owners' identities, financial statements and negotiation positions, so the design assumes an attacker wants to identify an anonymous practice or read a data room. The controls below answer the highest risks first; the India legal points at the end need a qualified lawyer before launch, and I am not giving legal advice.

Top threats and the control for each

Threat	Control
A buyer guesses IDs and reads another deal's records (broken object-level authorisation)	Policy functions check role, ownership, assignment and grant on every route; 404 instead of 403 for hidden records; automated tests for each route with the wrong user
An anonymous listing is re-identified from its details (city zone, chairs, revenue band and years combined)	Coarse revenue bands; zone not locality; no photos or opening year at Tier 0; a broker reviews every listing for identifiability before approval; rate limits and logging on listing views to catch scraping
Stolen or shared document link	60-second signed URLs, issued only after a permission check, with an access-log entry per issue
Malicious upload	Allow-listed types, content sniffing, 25 MB limit, malware scan before availability, files never executed or served from the app origin
Account takeover	bcrypt hashing, lockout after repeated failures, rotating refresh tokens with reuse detection, optional authenticator-app MFA required for brokers, managers and administrators (**recommended for MVP**)
Insider misuse	No standing admin access to documents; break-glass requires a reason and creates an audit entry that the brokerage manager is notified of
Injection and XSS	Zod validation on all input, Mongoose strict schemas, operator-injection stripping (`$` and `.` in keys), output encoding by React, a strict Content-Security-Policy
Denial of service and brute force	Redis-backed rate limits, request size limits, timeouts

Security controls by area

Area	Requirement
Transport	TLS 1.2 or higher everywhere; HSTS; secure cookies
At rest	Database and storage encryption at rest; application-level encryption of financials, exact address and practice name with keys in a key-management service; key rotation yearly
Secrets	Environment variables or a secrets manager; none in the repository; separate keys per environment
Sessions	15-minute access tokens; refresh tokens rotate on use; logout revokes the token family; sessions listed and revocable by the user
Audit	Append-only `auditLogs` written by a database user with insert-only rights; records logins, role changes, listing decisions, grants, revocations, views and downloads of confidential data
Retention (proposed, confirm with counsel)	Audit logs 7 years; closed-deal documents 7 years after closing, then deleted or returned; withdrawn-listing documents 12 months; unverified accounts 30 days; notifications 180 days
Backups	Daily encrypted snapshots, 30-day retention, quarterly restore test, backups in the same legal jurisdiction as primary data
Dependency and release	Dependency scanning and lockfile review in CI; no deploy with failing security tests

No patient records. The MVP stores no patient-identifiable information. Practice size may be stated only as an aggregate count (for example, active patients in a band); any upload containing patient names, such as an unredacted appointment export, is rejected during broker review, and the upload guidance says so.

India: what I could verify, and what is still open

- **DPDP Act.** The Digital Personal Data Protection Rules, 2025 were notified on 13 November 2025, with a phased rollout. Consent-manager provisions start about 12 months after notification, and the substantive obligations (notice, consent, security

safeguards, breach handling) are scheduled for 13 May 2027. The platform will be a data fiduciary for the personal data of owners, buyers and staff, so design for compliance from the start.

- **Possible acceleration.** MeitY proposed shortening some timelines in early 2026, and the Supreme Court has been notified of petitions challenging the Act and Rules. I could not confirm whether any change was adopted; **counsel should confirm the live dates**.
- **Design consequences:** a plain-language privacy notice at sign-up, purpose-specific consent records, a deletion and correction request flow, a breach-response runbook (the Rules reportedly set a 72-hour window for the detailed breach report; confirm), data minimisation (no Aadhaar numbers, no patient data), and a named contact for grievances.
- **Professional rules and advertising.** The Dentists (Code of Ethics) Regulations restrict dentists from advertising and allow announcements limited to matters such as a change of ownership. A listing that identifies a clinic could be read as advertising, which supports anonymised listings and is a reason to take legal advice before offering paid featured placement. A DCI advisory also tells state councils to act against violations of the Clinical Establishments Act, the Dentists Act and the Code of Ethics.

Questions for qualified legal and tax review (not answered here)

- Who may own a dental practice in India (dentist-only, or also companies and investors), and does this differ by state? One trade interview mentions a proposed National Dental Commission replacing the Dental Council of India; I could not confirm its current status.
- Which clinic registrations and licences transfer on a sale and which must be re-obtained (state registration, biomedical waste authorisation, radiation-equipment registration, local trade or establishment licences).
- Whether a platform that introduces buyers and sellers of share-based transactions needs any intermediary licence, and how brokerage fees are taxed (GST).
- Enforceability of clickwrap NDAs, and when an Aadhaar or digital-certificate e-signature is needed.
- Tax and stamp-duty treatment of asset versus share transfers, and of goodwill.
- The US, if you expand there: HIPAA applies only where the platform handles patient health information, which this design avoids.

20. AI Feature Strategy

No AI feature is necessary for the MVP: every MVP requirement is met by forms, rules and a database, and keeping AI out removes a data-protection and accuracy risk during the launch. AI is worth adding later in two places, drafting text and reading documents, and should stay out of anything that decides a price, a match or a legal position.

Candidate	Value	Risk	Verdict
Listing description drafting	Saves broker time	Low if it only receives non-identifying fields; a draft could still hint at identity	**P2, optional.** Broker edits and approves; a deterministic identifiability check runs after drafting
Financial document extraction (profit-and-loss, tax records to figures)	High: reduces data entry	Medium: misread numbers	**P3 pilot.** Show extracted values as "suggested, unverified" beside the source document; a broker confirms before anything is stored as verified
Document completeness and missing-document detection	High	Low	**Deterministic, P2.** A checklist by category, expiry dates and required-field rules; no AI needed
Buyer–practice matching	High	Medium: opaque scores damage trust	**Deterministic scoring, P2** (section 10F). Revisit learned weights only after you have real deal outcomes
Deal summaries for brokers (internal)	Medium	Low if internal only	**P3.** Read-only summary of timeline and notes
Due-diligence assistance (question lists, risk flags)	Medium	Medium	**P3, internal only.** Suggestions for brokers, never shown to sellers or buyers as findings

Candidate	Value	Risk	Verdict
Valuation number	–	High	**Not recommended.** The estimate stays a transparent formula; AI may later explain it in words but never produce it

Rules for any AI feature

- AI output is labelled as a suggestion, kept separate from verified facts in the data model (a `source` and `verifiedBy` field on every figure), and a person must approve it before it affects a listing, offer or decision.
- AI never makes acquisition, financial or legal decisions and never contacts anyone.
- Uploaded documents are untrusted: text inside a file is data to read, not instructions to follow, to block prompt injection.
- Confidential data goes only to a provider with a signed data-processing agreement, a no-training commitment and a data-residency position approved by counsel; sellers see in the privacy notice when AI touches their documents.
- Every AI call logs who triggered it, which document or record, and the model version, without storing the full confidential text in logs.
- Ship each feature with a small evaluation set (for example 30 real financial statements for extraction) and a target accuracy agreed before launch.

21. MVP Scope

The MVP is the smallest product that lets your brokerage team take a real dental practice from enquiry to closing inside the system, with confidentiality enforced by the server. It covers 35 features (section 13), and it is deliberately not a public marketplace.

In scope

Capability	Features
Accounts, roles, security	F01–F05, F40, F42, F43
Seller onboarding and sale request	F06–F08
Buyer onboarding and qualification	F09, F10
Listings: moderation, tiered anonymised view, search	F11–F14

Capability	Features
NDA, access control, logging	F17–F19
Documents and data room	F20–F22
Broker CRM and pipeline	F25–F30, F32, F34, F36
Notifications and administration	F37, F39, F41

Out of scope for the MVP

- Automated matching, the valuation estimator, full offer and counter-offer management, saved searches, in-app messaging, advisor roles, document comments and approval workflows (all Phase 2).
- E-signature, lender integration, external brokers, multi-organisation tenancy, any AI features (Phase 3 or later).
- Payment collection, patient records, native mobile apps, languages other than English.

Definition of done for the MVP release

1. A seller can submit a practice; a manager can approve it; a verified buyer sees only the anonymised view.
2. A buyer can request information, accept an NDA, and be approved; only then are identity and financials visible, and every view is logged.
3. A broker can move a deal across all pipeline stages with checklists and tasks.
4. All authorisation tests in section 23 pass, including the wrong-user test on every protected route.
5. A restore from backup has been tested once, and a legal review of the NDA text, privacy notice and terms is complete.

Launch constraints you must confirm: launch cities, whether brokers are all employees, and the NDA template wording (section 27).

22. Future Roadmap

Phase 2 starts only when the MVP has run real deals, and Phase 3 only when Phase 2 data justifies it; each phase has an exit gate so scope follows evidence rather than the calendar.

Phase	Scope	Gate to start
1. MVP	35 features in section 13	Legal sign-off on NDA, privacy notice and terms; launch cities confirmed
2. Advanced workflows	Automated matching (F35), valuation estimator (F33), due-diligence checklists (F24), document comments and approvals (F23), full offer management (F51), saved searches (F15), advisor access (F49), brokered messaging (F38), advanced analytics (F31), optional AI listing drafts	At least 10 deals run end-to-end in the MVP; valuation multiples validated against real Indian transactions; broker feedback reviewed
3. Scale	E-signature (F44), lender integration (F45), AI document extraction and internal summaries (F46), external brokers and multi-organisation support (F47), advanced reporting (F48), other specialties and cities	Phase 2 adoption metrics met; counsel confirms rules for external brokers and e-signature; security review passed

Sequencing rules

- Valuation ships only after its inputs are validated; a wrong early estimate damages trust more than having none.
- External brokers are not added until the audit, grant and revocation controls have run a full quarter without a finding.
- New specialties (veterinary, medical) reuse the same data model; only the specialty-specific fields and valuation configuration change.

23. Testing Strategy

Authorisation and confidentiality get the heaviest testing, because a single leak of an anonymous listing or a data-room file is the failure that would end trust in the product.

Layer	Tooling	What it covers	Gate
Unit	Vitest	Services, policy functions, Zod schemas, matching and valuation maths, tier-filtering functions	80% coverage on services and policies; all merges
API integration	Supertest with MongoDB Memory Server	Every route: success, validation failure, wrong role, wrong owner, expired grant, rate limit	Every route has at least one allow and one deny test
Database	Memory Server and a staging replica set	Unique and compound indexes, TTL indexes, transactions, soft delete, append-only audit user	Run in CI
Authentication	Supertest	Registration, verification expiry, lockout, reset-token reuse, refresh rotation and reuse detection, logout revocation	Run in CI
Authorisation matrix	Generated tests from the section 12 matrix	Each role against each protected action, plus wrong-user access to another user's records	Zero failures to release
Form validation	Vitest and Playwright	Same Zod schema in browser and API; error messages; multi-step draft saving	Run in CI
File upload	Supertest and fixtures	Allowed and blocked types, spoofed extension, 25 MB limit, EICAR test file, replace-with-version, expiry	Run in CI

Layer	Tooling	What it covers	Gate
Search and filtering	Supertest	Tier 0–1 fields only, filter combinations, sort whitelist, pagination edges, withdrawn listings absent	Run in CI
End-to-end	Playwright	The five journeys in the MVP definition of done	Before each release
Responsive and accessibility	Playwright at 360, 768, 1280 px; axe	No horizontal scroll; keyboard path through forms; no serious axe violations	Before each release
Security	Dependency audit, OWASP ZAP baseline scan, manual review	Injection, XSS, CSRF on refresh, IDOR, file upload abuse, header checks, scraping of anonymised listings	Before launch and after major changes
Performance	k6	Search and listing detail at the targets in section 11	Before launch

Edge cases that must have a test

- A buyer opens a listing URL that was withdrawn after they saved it: 404, and the saved item shows as unavailable.
- A grant expires mid-session: the next download request is refused.
- Two brokers move the same deal at once: the second gets a 409 and sees the new stage.
- A seller edits financial figures after a broker verified a document: the document returns to Pending.
- A buyer tries Tier 2 data by changing the ID in a URL, or by calling the API directly without the frontend.
- An uploaded file is renamed `.pdf` but is an executable: rejected on content type.
- A user's NDA acceptance refers to an older template version: acceptance is refused until the new version is accepted.

24. Acceptance Criteria

These criteria cover the critical MVP workflows in a testable given-when-then form; each maps to a feature ID in section 13 and to an automated test in section 23.

ID	Feature	Acceptance criterion
AC-01	F01, F02	Given a new user, when they register, then an account is created as unverified and a link valid for 24 hours is emailed; after verification, login succeeds; an expired or reused link returns 400 and does not verify.
AC-02	F02	Given 5 failed logins for one account within 15 minutes, when a 6th attempt is made even with the right password, then login is refused for 15 minutes and an audit entry is written.
AC-03	F02	Given a refresh token already used once, when it is presented again, then the request returns 401 and every token in that family is revoked.
AC-04	F06	Given a seller with a draft missing three mandatory fields, when they submit, then the API returns 422 naming all three fields and the status stays Draft; with all fields complete, status becomes Submitted and a lead is created for the brokerage.
AC-05	F11, F12	Given a submitted listing, when a manager approves it, then a `listings` document is generated containing only Tier 0 and Tier 1 fields; a search by any logged-in buyer returns it within 5 seconds; no response contains the name, address or exact financials.
AC-06	F12, F13	Given 50 approved listings, when a buyer searches by city and specialty with `pageSize=20`, then results contain only matching listings, `meta.total` is correct, and a request with `sort=revenue:asc` on a non-whitelisted field returns 400.
AC-07	F10, F17, F18	Given a buyer who has not accepted the NDA, when they request Tier 2 data, then the API returns 422 and no grant exists; after acceptance and broker approval, the grant exists with an expiry date and Tier 2 fields are returned.

ID	Feature	Acceptance criterion
AC-08	F18	Given an active grant, when a broker revokes it, then the buyer's next request for any Tier 2 field or document returns 404 within 1 second of the revocation.
AC-09	F20, F21	Given a PDF renamed from an executable, an EICAR test file and a 26 MB file, when each is uploaded, then each is rejected with 400 or 413 and none appears in storage; a clean 5 MB PDF becomes available only after the scan passes.
AC-10	F22, F19	Given a buyer with a document grant, when they request a download, then a URL valid for 60 seconds is returned and one access-log entry is written; the same URL fails after 60 seconds; a buyer without a grant receives 404.
AC-11	F29	Given a deal in Due diligence with an incomplete hard-gate checklist, when a broker moves it to Initial offer, then the API returns 422 with the missing items; a manager override with a reason succeeds and is audited.
AC-12	F03	Given the authorisation matrix in section 12, when each role calls each protected route, then every disallowed call returns 403 or 404 and every allowed call succeeds, with zero exceptions across the generated test suite.
AC-13	F27	Given a task due at 10:00 IST, when it is not completed by 10:00, then it appears in the assignee's overdue list and a reminder email is sent within 15 minutes.
AC-14	F41	Given any login, listing decision, grant, revocation, download or role change, when it occurs, then an audit entry with actor, action, entity and time exists and cannot be edited or deleted by any application user.
AC-15	F37	Given an NDA acceptance, when it is recorded, then the broker receives an in-app notification and an email within 2 minutes; the email contains no confidential figures or document names.

ID	Feature	Acceptance criterion
AC-16	F33 (Phase 2)	Given valuation inputs, when an estimate is produced, then the screen shows a low–mid–high range, every assumption and configuration version used, and the sentence stating it is not a certified valuation; the same inputs reproduce the same result.

25. Development Milestones

I estimate about 18 weeks from kickoff to a pilot with your brokerage team, assuming a team of two backend engineers, two frontend engineers, one part-time designer, one QA engineer and a product owner. This is a planning estimate, not a commitment; your team's size and the legal inputs on the critical path will move it. A Mermaid Gantt version is in the appendix.

Milestone	Weeks	Deliverable	Exit check
M0 Discovery and setup	1–2	Owner and buyer interviews started; launch cities, NDA wording and fee policy requested; repository, CI, environments, design tokens	Open questions in section 27 assigned to owners
M1 Foundations	3–5	Auth, sessions, RBAC policies, audit log, shared validation package, base components, role dashboards (shells)	AC-01 to AC-03, AC-12, AC-14 pass
M2 Onboarding and intake	6–8	Seller and buyer onboarding, sale request, financial intake, buyer profile and qualification	AC-04 passes; forms pass accessibility check
M3 Listings and discovery	9–10	Moderation queue, anonymised tier generation, search and filters, interests	AC-05, AC-06 pass

Milestone	Weeks	Deliverable	Exit check
M4 NDA, access and data room	11–12	NDA templates and acceptance, grants and revocation, private uploads, malware scan, signed downloads, access log	AC-07 to AC-10 pass; legal has approved the NDA
M5 Broker CRM and administration	13–15	Leads, assignment, tasks, notes, Kanban pipeline with gates, notifications, admin dashboard	AC-11, AC-13, AC-15 pass
M6 Hardening	16–17	Security scan and manual review, load test, accessibility and responsive audit, backup restore drill, copy and empty-state polish	No critical or high findings open
M7 Pilot launch	18	Brokerage team runs 3–5 live practices; daily review of issues	Definition of done in section 21 met

Critical path: legal sign-off on the NDA and privacy notice must be ready by week 11, and launch cities by week 6, or M4 and M2 slip. Phase 2 planning starts after week 18 using what the pilot teaches.

26. Risks and Mitigation

The two risks most likely to hurt the launch are unclear legal rules on dental practice ownership and listing promotion, and a thin supply of sellers willing to list; both are business risks rather than technical ones.

#	Risk	Likelihood	Impact	Mitigation	Owner
R1	Ownership, licence-transfer or advertising rules restrict how practices can be sold or promoted	Medium	High	Legal review before launch (section 19 questions); anonymised listings by default; no paid featured placement until cleared	Founder, counsel

#	Risk	Likelihood	Impact	Mitigation	Owner
R2	Too few sellers to make discovery worthwhile	High	High	Start broker-led with your existing owner relationships; treat the platform as the service, not a classifieds site; measure listings approved per month	Founder
R3	An anonymous listing is re-identified	Medium	High	Coarse bands, broker identifiability review, no photos before NDA, scrape monitoring	Product, security
R4	Confidential document leak	Low	Very high	Signed URLs, grant checks, audit logs, no standing admin access, penetration test before launch	Engineering
R5	DPDP obligations arrive earlier than planned or are read more strictly	Medium	Medium	Build consent, notice, deletion and breach processes into the MVP; counsel confirms live dates	Counsel, engineering
R6	Indian financial records are informal, so verification is slow	High	Medium	Guided intake, document checklist, broker verification step; set expectations in the seller journey	Brokerage
R7	Valuation estimate is wrong and damages trust	Medium	High	Delay to Phase 2; validate multiples with real deals; show assumptions; never present as certified	Product
R8	Scope creep from Phase 2 features	High	Medium	Section 13 gates; change requests need a founder decision and a dropped feature	Product owner
R9	Brokers keep working in spreadsheets	Medium	High	Involve two brokers in design from week 1; import tool for existing leads; track share of deals in the system	Brokerage manager
R10	Competitor comparisons were made from homepages only	Certain	Medium	Request demos from Dental Acquisitions and a broker-led US firm; update sections 4 and 5	Product
R11	Single-region hosting outage	Low	Medium	Daily backups, tested restore, documented recovery steps	Engineering

27. Assumptions and Open Questions

Three facts are confirmed by you; everything else below is my assumption or a decision only you can make.

Confirmed

Item	Answer
Operating geography	India first
Who runs transactions	Your own brokerage team (internal CRM-led)
MVP vertical	Dental only

Assumptions I made (change any of them)

#	Assumption	Effect if wrong
A1	Brokers and managers are all your employees; external brokers wait for Phase 3	Needs multi-organisation permissions earlier
A2	Sellers and buyers are individuals or small entities, not large groups	Group acquirers need an organisation workflow
A3	English-only interface at launch	Hindi or regional languages add translation and layout work
A4	Signing of NDAs is by clickwrap, with e-signature in Phase 3	Earlier e-sign integration adds about 3 weeks
A5	File limit of 25 MB and the listed document types	Larger scans need chunked upload
A6	Team of about six people and an 18-week timeline	Smaller team lengthens the plan roughly in proportion
A7	Valuation estimator waits until Phase 2 because no validated Indian multiples exist	If you have internal pricing data, it can move earlier
A8	MFA is required for staff accounts	Dropping it reduces account-takeover protection

Decisions needing founder approval

1. Launch cities, and whether to start in one city.

2. Fee model and rate, and whether your team represents sellers only or both parties (section 7).
3. Whether only your brokerage can approve listings at launch (section 6).
4. Whether sellers may show an asking price at Tier 0 or only after the NDA.
5. Whether buyers can ever message sellers directly, or all contact stays broker-mediated.
6. Hosting provider and cloud region, once counsel has advised on data residency.
7. Whether to offer paid featured placement, which needs legal review first (section 19).
8. The brand name and domain.

Questions for the business team

- Do you already have sellers and buyers, and how many, to seed the platform?
- Can you share 5–10 anonymised past deals for validating valuation and the pipeline stages?
- Which document types do sellers in your cities usually have, and how reliable is their financial record-keeping?
- Does any lender or accountant partner need access to deals?

Information I could not verify: Indian dental transaction volumes and multiples, competitor features behind logins, and the live status of DPDP timelines and the National Dental Commission.

28. Appendices

Diagrams are given as Mermaid source so your team can paste them into any Mermaid viewer, wiki or the repository's `docs/` folder. The feature prioritisation table is section 13 and the MVP timeline is section 25.

A. Diagrams

A1. System architecture

```
flowchart LR
  U["Browser: seller, buyer, broker, admin"] -->|HTTPS| W["Next.js web app"]
  W -->|"REST /api/v1 + access token"| A["Express API"]
  A --> P["Policy layer: role, ownership, grant"]
  A --> M[("MongoDB replica set")]
  A --> R[("Redis")]
  R --> J["Job worker: malware scan, email, reminders"]
  A -->|"60 s signed URL"| S[("Private object storage")]
  U -->|"download via signed URL"| S
  J --> S
  J --> E["Email provider"]
  A --> L["Logs and error tracking"]
```

A2. Disclosure workflow: from interest to data room

```
flowchart TD
  A["Buyer sees Tier 0-1 listing"] --> B["Buyer expresses interest"]
  B --> C{"Profile complete and broker-qualified?"}
  C -- No --> C1["Broker qualifies or declines with note"]
  C1 --> C
  C -- Yes --> D["Buyer accepts current NDA version"]
  D --> E["Broker reviews access request"]
  E --> F{"Seller veto?"}
  F -- Yes --> G["Request rejected, buyer notified"]
  F -- No --> H["Grant created with scope and expiry"]
  H --> I["Tier 2 fields and approved documents visible"]
  I --> J["Every view and download logged"]
  I --> K{"Expired or revoked?"}
  K -- Yes --> L["Access removed on next request"]
```

A3. Transaction lifecycle

```
stateDiagram-v2
  [*] --> NewLead
  NewLead --> BuyerQualified
  BuyerQualified --> NDA
  NDA --> InformationDisclosure
  InformationDisclosure --> DueDiligence
  DueDiligence --> InitialOffer
  InitialOffer --> Negotiation
  Negotiation --> LetterOfIntent
  LetterOfIntent --> LegalReview
  LegalReview --> FinalAgreement
  FinalAgreement --> Closing
  Closing --> OwnershipTransition
  OwnershipTransition --> PostSaleHandover
```

```
  PostSaleHandover --> [*]
  NewLead --> Lost
  BuyerQualified --> Lost
  DueDiligence --> Lost
  Negotiation --> Lost
  Negotiation --> Withdrawn
  Lost --> [*]
  Withdrawn --> [*]
```

A4. Listing approval states

```
stateDiagram-v2
  [*] --> Draft
  Draft --> Submitted
  Submitted --> InReview
  InReview --> ChangesRequested
  ChangesRequested --> Submitted
  InReview --> Rejected
  InReview --> Approved
  Approved --> Listed
  Listed --> UnderOffer
  UnderOffer --> Sold
  Listed --> Withdrawn
  UnderOffer --> Withdrawn
```

A5. MongoDB collection relationships

```
erDiagram
  USERS ||--o| BUYERPROFILES : has
  USERS ||--o| SELLERPROFILES : has
  ORGANIZATIONS ||--o{ USERS : employs
  USERS ||--o{ PRACTICES : owns
  PRACTICES ||--o{ PRACTICEFINANCIALS : has
  PRACTICES ||--o| LISTINGS : publishes
  PRACTICES ||--o{ DOCUMENTS : holds
  LISTINGS ||--o{ BUYERINTERESTS : receives
  USERS ||--o{ BUYERINTERESTS : makes
  USERS ||--o{ NDAACCEPTANCES : signs
  NDATEMPLATES ||--o{ NDAACCEPTANCES : version
  USERS ||--o{ ACCESSGRANTS : receives
  LISTINGS ||--o{ ACCESSGRANTS : scopes
  PRACTICES ||--o{ DEALS : sells
  DEALS }o--|| PIPELINESTAGES : at
  DEALS ||--o{ TASKS : has
```

```
  LEADS ||--o{ TASKS : has
  DEALS ||--o{ ACTIVITIES : logs
  USERS ||--o{ NOTIFICATIONS : gets
  USERS ||--o{ AUDITLOGS : acts
```

A6. MVP timeline (weeks)

```
gantt
  dateFormat  X
  axisFormat  W%s
  section Setup
  M0 Discovery and setup      :0, 2
  section Build
  M1 Foundations              :2, 5
  M2 Onboarding and intake    :5, 8
  M3 Listings and discovery   :8, 10
  M4 NDA, access, data room   :10, 12
  M5 Broker CRM and admin     :12, 15
  section Release
  M6 Hardening                :15, 17
  M7 Pilot launch             :17, 18
```

B. Initial API endpoint inventory

Roles: Pub public, Any any signed-in user, SE seller, BU buyer or investor, BR assigned broker, BM manager, AD super or platform admin. Every route also applies the ownership, assignment or grant checks in section 12.

Group	Endpoints (method path, roles)
Auth (9)	POST /auth/register (Pub); POST /auth/login (Pub); POST /auth/refresh (cookie); POST /auth/logout (Any); POST /auth/verify-email (Pub); POST /auth/resend-verification (Pub); POST /auth/forgot-password (Pub); POST /auth/reset-password (Pub); GET /auth/me (Any)
Users and profiles (9)	GET, PATCH /users/me (Any); POST /organizations (Any); GET, PATCH /organizations/:id (members, AD); GET, PUT /profiles/buyer/me (BU); GET, PUT /profiles/seller/me (SE)

Group	Endpoints (method path, roles)
Verification (4)	POST /verification-requests (SE, BU); GET /verification-requests/me (SE, BU); GET /admin/verification-requests (AD); PATCH /admin/verification-requests/:id (AD)
Practices (8)	POST /practices (SE, BR); GET /practices (SE own, BR assigned, BM); GET, PATCH /practices/:id (owner, BR, BM); POST /practices/:id/submit (SE); POST /practices/:id/withdraw (SE, BM); PUT /practices/:id/financials/:fiscalYear (SE, BR); GET /practices/:id/financials (owner, BR, BM, Tier 2 grant)
Listings (8)	GET /public/teasers (Pub; at most 6 cached Tier 0 cards, no search); GET /listings (BU, BR, BM, AD; Tier 0–1); GET /listings/:id (tiered); POST, DELETE /listings/:id/favourite (BU); GET /favourites (BU); GET /admin/listings/queue (BM, AD); POST /admin/listings/:practiceId/decision (BM, AD: approve, request changes, reject)
Interests, NDA, access (13)	POST /listings/:id/interests (BU); GET /interests/me (BU); DELETE /interests/:id (BU); GET /broker/interests (BR, BM); PATCH /buyers/:id/qualification (BR, BM); GET /buyers (BR, BM); GET /nda/templates/current (BU); POST /nda/accept (BU); GET /access-requests (scoped by role); POST /access-requests/:id/approve (BR, BM); POST /access-requests/:id/reject (BR, BM); POST /access-requests/:id/veto (SE); POST /access-grants/:id/revoke (BR, BM, SE)
Documents (7)	POST /documents/uploads (SE, BR, BU own); POST /documents/:id/complete (uploader); GET /documents (by practice or deal, scoped); GET /documents/:id (scoped); GET /documents/:id/download-url (owner, BR, BM, grant); PATCH /documents/:id/review (BR, BM); DELETE /documents/:id (uploader before review, BM)

Group	Endpoints (method path, roles)
CRM (11)	POST, GET /leads (BR, BM); GET, PATCH /leads/:id (BR, BM); POST /leads/:id/assign (BM); POST /leads/:id/convert (BR, BM); POST, GET /tasks (BR, BM); PATCH /tasks/:id (assignee, BM); POST /activities (BR, BM); GET /activities? entityType&entityId (BR, BM)
Deals and pipeline (12)	POST, GET /deals (BR, BM); GET, PATCH /deals/:id (BR, BM; seller sees own stage); PATCH /deals/:id/stage (BR, BM); POST /deals/:id/offers (BR, BM; basic record); GET /deals/:id/timeline (BR, BM); GET /pipeline/stages (BR, BM); POST, PATCH /admin/pipeline/stages (BM, AD); GET, PUT /listings/:id/shortlist (BR, BM)
Notifications (4)	GET /notifications (Any); PATCH /notifications/:id/read (Any); POST /notifications/read-all (Any); PUT /notifications/preferences (Any)
Admin and analytics (11)	GET /admin/dashboard (AD); GET /admin/users (AD); POST /admin/users/invite (AD); PATCH /admin/users/:id (AD); GET /admin/audit-logs (AD; BM own brokerage); GET, POST /admin/nda-templates (AD); PATCH /admin/nda-templates/:id (AD); POST /admin/break-glass (SA only, reason required); GET /analytics/broker (BR); GET /analytics/brokerage (BM)
Platform (1)	GET /health (Pub, no data)

C. Technical decision log

ID	Decision	Alternative considered	Reason
TD-01	pnpm monorepo with a shared Zod package	Two separate repositories	One validation source; atomic changes across web and API
TD-02	Separate Express API; no Next.js API routes	Next.js route handlers	Mandated by your stack; keeps security logic out of the web tier

ID	Decision	Alternative considered	Reason
TD-03	Private object storage with signed URLs	Cloudinary	Confidential documents need private buckets and access logs
TD-04	Access token in header, refresh token in httpOnly cookie	Session cookie only	Limits CSRF to one endpoint; lets staff sessions be revoked
TD-05	Anonymised listing stored as its own document	Filtering fields at read time	A coding mistake cannot leak private fields from a document that never holds them
TD-06	Application-level encryption of financials, address and name	Encryption at rest only	Protects against database-level access and backups
TD-07	404 instead of 403 for hidden records	403 everywhere	Does not reveal that a confidential record exists
TD-08	Queue-based malware scanning	Synchronous scan in the request	Keeps uploads fast and the API responsive
TD-09	Clickwrap NDA with stored hash, time and IP in MVP	Third-party e-signature now	Faster launch; e-signature in Phase 3 after legal advice
TD-10	Deterministic matching and valuation	AI-generated scores	Transparent, testable, defensible to sellers
TD-11	Revenue shown as bands before NDA	Exact revenue	Reduces re-identification risk
TD-12	Staff MFA required	Optional MFA	Staff hold the most sensitive access

D. Review findings (step 9)

I reviewed the PRD for contradictions, security gaps and unnecessary complexity. Findings and how each was handled:

- **Fixed:** section 10D said any visitor can see Tier 0 teasers, while `GET /listings` requires sign-in. Resolution: a separate cached `GET /public/teasers` endpoint (maximum six cards, no search) serves the landing page.

- **Fixed:** the access approval endpoint returns an error for a seller veto, but no route recorded a veto. Added `POST /access-requests/:id/veto`.
- **Fixed:** a feature count of 36 in an earlier draft was wrong; section 13 now states 35 MVP features, 31 Must-haves.
- **Open:** the 25 MB limit, retention periods and rate limits are proposals, not researched standards.
- **Open:** competitor findings rest on homepages; section 4 labels this, and R10 schedules demo requests.
- **Simplified:** offers are a basic record in MVP, with full offer management deferred to Phase 2 to keep the first release small.

E. Implementation-readiness checklist

- [ ] Founder decisions 1–8 in section 27 answered
- [ ] Launch cities confirmed
- [ ] NDA wording, privacy notice and terms approved by counsel
- [ ] Legal answers on ownership, licence transfer and advertising rules received
- [ ] Hosting provider and region chosen, with data-residency advice
- [ ] Two brokers and two owners interviewed; pipeline stages and document checklist confirmed
- [ ] Design tokens and component library agreed (section 15)
- [ ] Repository, CI and environments created; secrets manager in place
- [ ] Authorisation test suite generated from section 12 and wired into CI
- [ ] Backup and restore procedure written
- [ ] Pilot success criteria and review cadence agreed